

Residential House Price Prediction – Univariate and Bivariate Analysis

Objective:

The objective of this phase is to explore and understand the characteristics of the residential house dataset using statistical analysis and graphical visualization techniques. Univariate analysis was performed to study individual feature distributions, while bivariate analysis was conducted to identify relationships between variables and their influence on house sale prices.

1. Univariate Analysis:

Overview:

Univariate analysis focuses on examining each variable independently to understand its distribution, central tendency, variability, and data characteristics.

The analysis was performed using descriptive statistical measures and visualization techniques.

Descriptive Statistical Analysis:

We analyzed the data using statistics to understand its shape, average values, variation, and patterns, and we used a Python tool to automate this process.

Statistical measures were generated for numerical variables to summarize the dataset.

The following measures were calculated:

- Mean
- Median
- Mode
- First Quartile (Q1 – 25%)
- Second Quartile (Q2 – 50%)
- Third Quartile (Q3 – 75%)
- 99th Percentile
- Maximum Value
- Interquartile Range (IQR)
- Variance
- Standard Deviation
- Kurtosis
- Skewness

These metrics help identify:

- Data distribution

- Central tendency
- Spread of observations
- Presence of skewness
- Variability across features

The descriptive statistics were generated using a reusable Python analysis module.

Frequency Distribution Analysis:

Frequency analysis was performed for categorical variables to understand the occurrence of each category.

The following metrics were generated:

- Frequency
- Relative Frequency
- Cumulative Frequency

This analysis helped identify dominant categories and feature distributions.

Probability Distribution Analysis

Probability Density Function (PDF) analysis was performed to evaluate the probability distribution of selected numerical features.

The analysis included:

- Mean calculation
- Standard deviation calculation
- Distribution visualization
- Probability estimation within a selected range

This helped understand how observations are distributed across the dataset.

Univariate Visualizations

Histogram

Histogram plots were generated to analyze the distribution pattern of numerical variables.

Purpose:

- Identify skewness
- Observe distribution shape
- Detect concentration of values

Distribution Plot

Distribution plots with Kernel Density Estimation (KDE) were generated to visualize smooth probability distributions.

Purpose:

- Understand data spread
- Observe distribution density

Box Plot

Box plots were generated to examine data spread and detect outliers.

Purpose:

- Visualize quartiles
- Identify extreme values
- Evaluate data consistency

Count Plot

Count plots were used for categorical variables.

Purpose:

- Visualize category frequencies
- Compare category distribution

2. Bivariate Analysis

Overview

Bivariate analysis was conducted to study relationships between independent variables and the target variable (SalePrice).

This analysis helps identify features that strongly influence house price prediction.

Correlation Analysis

Correlation analysis was performed to measure the strength and direction of relationships between numerical variables.

Correlation values range between:

- +1 → Strong positive relationship

- $0 \rightarrow$ No relationship
- $-1 \rightarrow$ Strong negative relationship

This analysis supports feature selection during model building.

Covariance Analysis

Covariance analysis was performed to determine whether variables move together.

Purpose:

- Measure joint variability
 - Understand directional relationships
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Variance Inflation Factor (VIF)

VIF analysis was performed to detect multicollinearity among numerical features.

Interpretation:

- $VIF < 5 \rightarrow$ Acceptable
- $VIF > 5 \rightarrow$ Moderate correlation
- $VIF > 10 \rightarrow$ High multicollinearity

This step helps improve model stability.

Bivariate Visualizations:

Scatter Plot

Scatter plots were generated to examine relationships between two numerical variables.

Examples:

- Living Area vs Sale Price
- Garage Area vs Sale Price

Purpose:

- Identify trends
 - Detect patterns
 - Observe linear relationships
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Joint Plot

Joint plots were used to visualize both variable relationships and individual distributions simultaneously.

Purpose:

- Relationship analysis
 - Distribution comparison
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Pair Plot

Pair plots were generated to compare relationships among multiple numerical features.

Purpose:

- Detect patterns
 - Explore feature interactions
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Correlation Heatmap

Heatmaps were generated to visualize correlations between numerical variables.

Purpose:

- Identify strong relationships
 - Support feature selection
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Conclusion:

The Univariate and Bivariate analysis phase provided a comprehensive understanding of feature distributions and inter-variable relationships within the residential house dataset. Statistical measures and visualization techniques helped identify important patterns, understand feature behavior, and support the upcoming stages of feature engineering and predictive model development.
